

MTH209 Assignment 2, IIT Kanpur

14 Jan 2026 | Time : 15 minutes | Total Points : 10

Q1. Consider matrices $A \in \mathbb{R}^{1000 \times 100}$ and $B \in \mathbb{R}^{100 \times 100}$ and a vector $c \in \mathbb{R}^{100}$. We want to calculate ABc . Which of the following has lower computational complexity (if the complexity is the same, select both):

- $(AB)c$
- ✓ $A(Bc)$

Q2. The matrix

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive-definite.

- ✓ **True**
- False

Q3. Which of the following is generally the fastest way to invert a 1000×1000 positive semi-definite matrix?

- Singular value decomposition
- ✓ **Cholesky decomposition**
- QR decomposition
- Inversion by formula (adjoint and determinant)

Q4. Consider samples $X_1, X_2, \dots, X_n$ from some one-dimensional distribution. Calculation of sample variance is of what complexity (in order)?

Score: 1 Neg. Score: 0

- $O(\sqrt{n})$
- $O(n^3)$
- ✓ $O(n)$
- $O(n^2)$

Q5. Cholesky decompositions hold for all matrices

- True
- ✓ **False**

Q6. Consider a vector $v \in \mathbb{R}^{100}$. Then the matrix $vv^\top$

- has rank 100
- is a positive definite matrix that can never be singular
- ✓ is a positive semi-definite matrix
- has both positive and negative eigenvalues
- has determinant equal to 1

Q7. The command: `R <- chol(A)` is applied on a positive definite matrix A . Then

- R is a lower triangular matrix
- ✓ R is an upper triangular matrix
- R is a diagonal matrix
- R is the inverse of matrix A

Q8. Consider the least squares estimator $(X^\top X)^{-1}(X^\top y)$ where $X \in \mathbb{R}^{n \times p}$ and $y \in \mathbb{R}^n$. What is the computational complexity of this estimator, as a function of both p and n , using direct multiplication?

- ✓ $O(p^3 + p^2n)$
- $O(p^3)$
- $O(n^3 + p^2)$
- $O(n^3 + p^2n)$

Q9. The command `"svd.A <- svd(A)"` implements the singular value decomposition of a matrix A . If A is an 1000×100 matrix of rank 50, then the object `"svd.A$d"` is expected to be a

- ✓ numeric vector of size 100
- numeric vector of size 50
- diagonal matrix of size 100×100
- diagonal matrix of size 50×50

Q10. In the code below A is a large square matrix.

```
solve.A.inv <- solve(A)
qr.A.inv <- qr.solve(A)
```

Irrespective of A , `solve.A.inv` and `qr.A.inv` will always be exactly the same.

- True
- ✓ False